

Assumptions of Linear Regression ● Intermediate

Five conditions the model quietly assumes are true — and what happens when they aren't.

1. Linearity

What it means: the true relationship between features and target is approximately a straight line (or plane, with multiple features).

Why it matters: if the real relationship curves, a straight line systematically mis-predicts in certain ranges.

How to detect it: plot residuals against predicted values — a clear curve pattern signals non-linearity.

What it looks like: a residual plot with a visible U-shape or arc instead of random scatter.

What to do: add polynomial terms (Chapter 22), transform a feature (e.g. log), or use a non-linear model.

2. Independence

What it means: one observation's error shouldn't predict another's — no hidden ordering or repetition effect.

Why it matters: violated independence (common in time series) makes standard errors and significance tests unreliable, even if predictions still look fine.

How to detect it: plot residuals in the order they occurred; look for runs or cycles.

What it looks like: residuals that snake up and down together in sequence, rather than jumping randomly.

What to do: use models designed for sequential data, or add lag features.

3. Homoscedasticity

What it means: the spread of residuals stays roughly constant across all predicted values.

Why it matters: if error spread changes (heteroscedasticity), the model is more uncertain in some ranges than others, which standard error estimates don't account for.

How to detect it: the residual-vs-predicted plot again — this time looking at spread, not shape.

What it looks like: a funnel or cone shape, where residuals fan out as predictions grow.

What to do: transform the target (e.g. log of y), or use weighted least squares.

4. Normality of residuals

What it means: residuals are approximately normally distributed.

Why it matters: mainly affects the validity of confidence intervals and hypothesis tests on coefficients — not the point predictions themselves.

How to detect it: a histogram of residuals, or a Q-Q plot.

What it looks like: a lopsided or heavy-tailed residual histogram instead of a roughly bell-shaped one.

What to do: often tolerable in practice with enough data; consider transformations or robust methods if severe.

5. No severe multicollinearity (multiple regression only)

What it means: predictor variables shouldn't be too strongly correlated with each other.

Why it matters: makes individual coefficient estimates unstable and hard to interpret (Chapter 18 goes deep here).

How to detect it: correlation matrix between features, or VIF.

What it looks like: coefficients that swing wildly or flip sign when you add/remove a similar feature.

What to do: remove or combine redundant features, or regularize.

DON'T MEMORIZE

Don't memorize assumptions as a list to recite. Understand what each one looks like in a residual plot — that's the version of this knowledge you'll actually use.

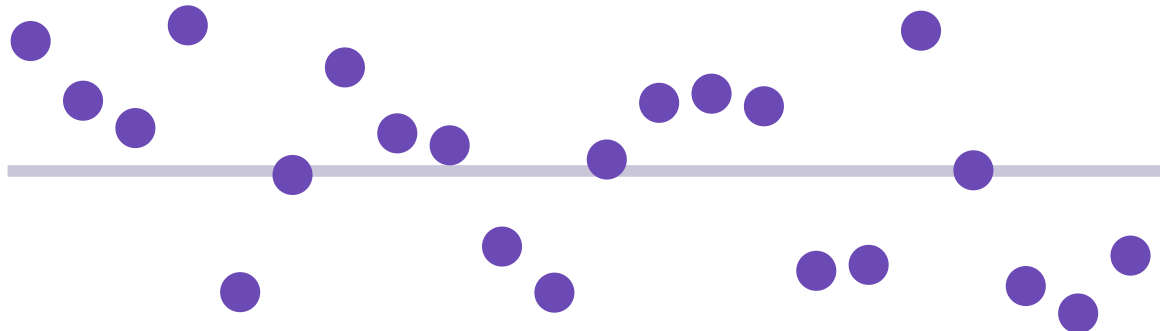
Residual Analysis ● Intermediate

The single most under-used diagnostic tool in applied regression.

DEFINITION, AGAIN – WORTH REPEATING

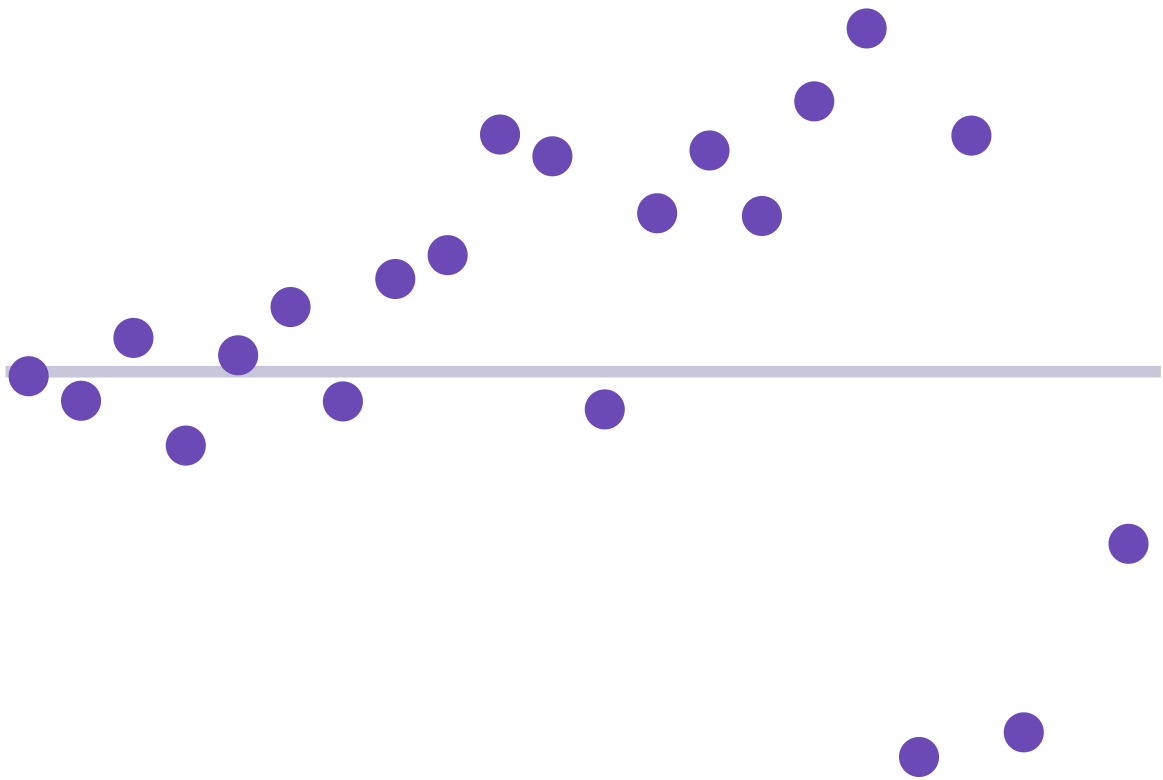
Residual = Actual – Predicted

Residual vs. Predicted plot — reading the shapes

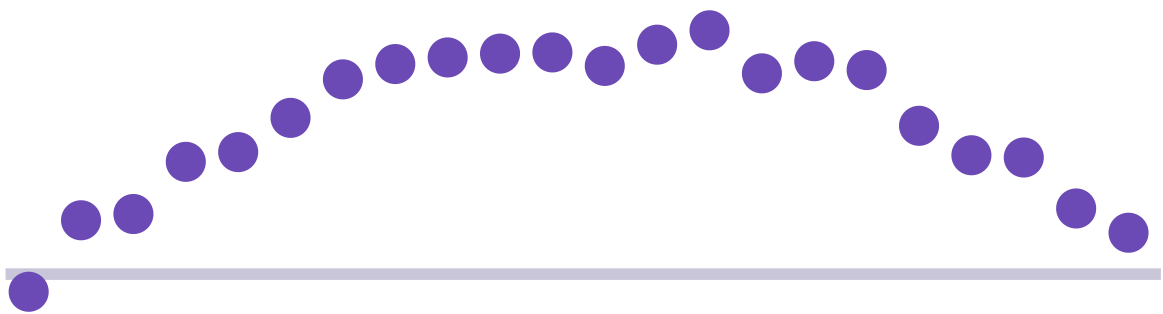


Random scatter

Healthy — no visible pattern



Funnel shape
Heteroscedasticity



Curve pattern
Non-linearity missed

COMMON MISTAKE

Looking only at R^2 or RMSE and skipping the residual plot entirely. A model can have a "good" overall RMSE while badly mispredicting an entire region of the data — the kind of systematic error a single number will never reveal, but a residual plot exposes instantly.

Multicollinearity Advanced

What happens when your predictors start predicting each other.

EXAMPLE

House size, number of rooms, and number of bedrooms are all highly correlated with each other — a bigger house tends to have more rooms *and* more bedrooms. Including all three makes it hard for the model to tell which one is "really" driving price.

WHY THIS CREATES UNSTABLE COEFFICIENTS

When two features carry almost the same information, the model can trade coefficient weight between them in many different ways and land on nearly the same predictions — so small changes in the data can swing individual coefficients dramatically, even flipping their sign, without changing overall accuracy.

VIF — VARIANCE INFLATION FACTOR

Conceptually, VIF measures how much a feature's variance is "inflated" because it can be predicted from the other features. A high VIF flags a feature that's largely redundant with the rest of the feature set. Don't over-focus on memorizing exact thresholds — the point is the direction and relative severity.

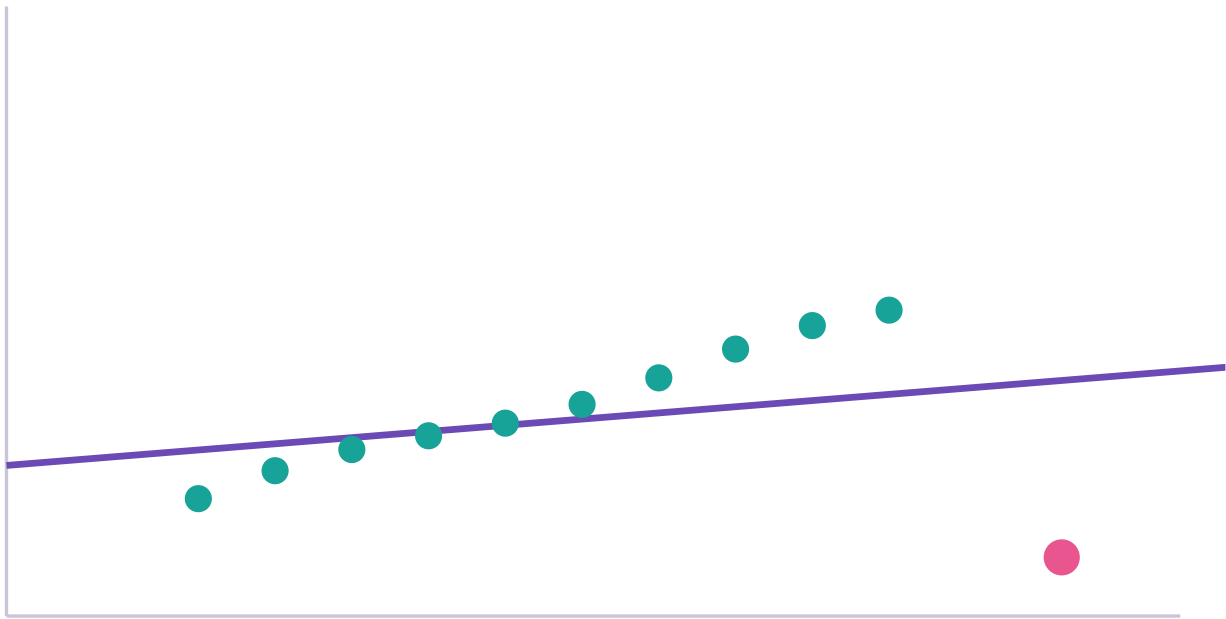
Common approaches once multicollinearity is detected: remove one of the redundant variables, combine them into a single composite feature, apply regularization (Chapter 23), or lean on domain knowledge to choose which variable is more meaningful to keep.

Outliers and Influential Points ● Intermediate

One unusual point can move the entire model — see it happen below.

Term	Meaning
Outlier	A point far from the general trend of the data (in x, y, or both)
Leverage	How unusual a point's <i>feature</i> value is, independent of its target value
Influential observation	A point (often high-leverage) whose removal would noticeably change the fitted line

Interactive — Outlier Influence



Line without outlier vs with: slope shifts from 0.88 to 0.26

Drag the highlighted pink point far from the rest of the cluster and watch the regression line tilt toward it.

WHY LINEAR REGRESSION IS SENSITIVE TO OUTLIERS

Because errors are squared, a point that's far from the line contributes a disproportionately huge amount to SSE — so OLS will happily distort the whole line to shrink that one large squared term, even at the cost of fitting everything else slightly worse.

At a high level, robust alternatives exist (e.g. Huber loss, RANSAC, quantile regression) that reduce the influence of extreme points — worth knowing exist, even if plain OLS remains the default starting point.